

KELT-23Ab

R-2138

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-23A_250526 · created 2026-08-12T00:02:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460821.92237 +/- 0.00066 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1311 +/- 0.0042
Transit depth (Rp/Rs)^2	1.72 +/- 0.11 [%]
Orbital Inclination (inc)	86.17 +/- 0.46
Airmass coefficient 1 (a1)	1.174 +/- 0.0056
Airmass coefficient 2 (a2)	-0.1233 +/- 0.005
Scatter in the residuals of the lightcurve fit is	0.48 %
Best Comparison Star	None
Optimal Aperture	3.64
Optimal Annulus	6.28
Transit Duration (day)	0.0955 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1311 ± 0.0042 vs 0.132 (deviation 0.18σ)
Duration fit vs published	0.0955 d vs 0.0954 d (deviation 0.02σ)
Mid-time O–C	-2.3 min
Depth SNR	26.2
Residual scatter	0.48 %

Light curve

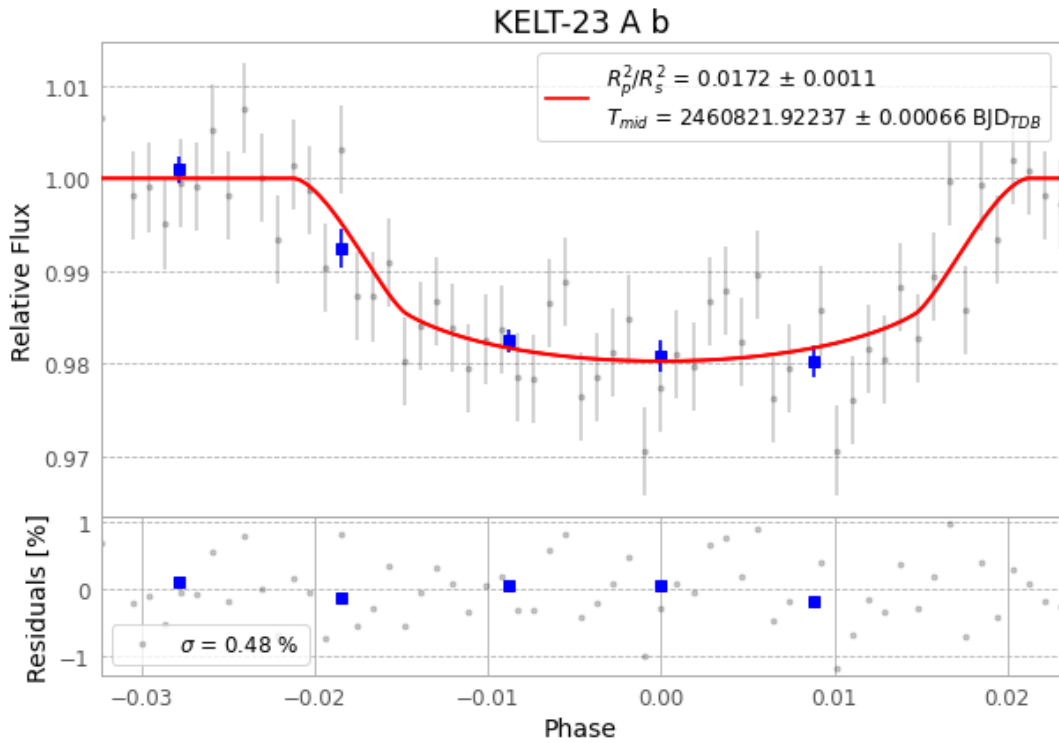


Figure 1 — FinalLightCurve_KELT-23 A b_2025-05-26.png (SHA-256 ecff6a30442423d3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 286], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 069fe954bcde9101...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

61 FITS frames (40 MB), KELT-23A250526082115.FITS → KELT-23A250526112115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-23A-250526.log (c1e2759d86d0...), FinalLightCurve_KELT-23 A b_2025-05-26.png (ecff6a304424...), FinalLightCurve_KELT-23 A b_2025-05-26.csv (ee1d4f9133b4...)

Compute resources

Wall time 167.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:02Z.